

DS LAB 8

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TASK NO 1

```cpp
#include <iostream>
using namespace std;
int binarySearch(int arr[], int size, int key)
{
    int low = 0;
    int high = size - 1;

    while(low <= high)
    {
        int mid = (low + high) / 2;

        if(arr[mid] == key)
            return mid;

        if(key > arr[mid])
            low = mid + 1;
        else
            high = mid - 1;
    }

    return -1;
}
void bubbleSort(int arr[], int size)
{
    for(int i = 0; i < size - 1; i++)
    {
        for(int j = 0; j < size - 1 - i; j++)
        {
            if(arr[j] > arr[j + 1])
            {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
void insertionSort(int arr[], int size)
{
    for(int i = 1; i < size; i++)
    {
        int key = arr[i];
        int j = i - 1;

        while(j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }
}
```

```cpp
// Function for Selection Sort
void selectionSort(int arr[], int size)
{
    for(int i = 0; i < size - 1; i++)
    {
        int minIndex = i;

        for(int j = i + 1; j < size; j++)
        {
            if(arr[j] < arr[minIndex])
                minIndex = j;
        }

        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }
}

// Function to print the array
void printArray(int arr[], int size)
{
    for(int i = 0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main()
{
    int const SIZE = 10;
    int arr[SIZE];
    int choice;
    bool isSorted = false; // Track whether array is sorted for binary search

    // 1. Take 10 integers input from the user
    cout << "Enter " << SIZE << " integers for the array:\n";
    for(int i = 0; i < SIZE; i++)
    {
        cout << "Element " << i + 1 << ": ";
        cin >> arr[i];
    }

    do {
        cout << "\n============================\n";
        cout << "        MENU-DRIVEN PROGRAM     \n";
        cout << "============================\n";
        cout << "1. Bubble Sort\n";
        cout << "2. Insertion Sort\n";
        cout << "3. Selection Sort\n";
        cout << "4. Binary Search\n";
```

```
    cout << "5. Display Array\n";
    cout << "6. Exit\n";
    cout << "Enter your choice (1-6): ";
    cin >> choice;

    switch(choice)
    {
        case 1:
            bubbleSort(arr, SIZE);
            isSorted = true;
            cout << "\nArray sorted successfully using Bubble Sort!\n";
            cout << "Sorted array: ";
            printArray(arr, SIZE);
            break;

        case 2:
            insertionSort(arr, SIZE);
            isSorted = true;
            cout << "\nArray sorted successfully using Insertion Sort!\n";
            cout << "Sorted array: ";
            printArray(arr, SIZE);
            break;

        case 3:
            selectionSort(arr, SIZE);
            isSorted = true;
            cout << "\nArray sorted successfully using Selection Sort!\n";
            cout << "Sorted array: ";
            printArray(arr, SIZE);
            break;

        case 4:
        {
            if (!isSorted) {
                cout << "\n[Note]: Binary search requires a sorted array. Sorting array automatically with Bubble Sort first...\n";
                bubbleSort(arr, SIZE);
                isSorted = true;
                cout << "Array now sorted: ";
                printArray(arr, SIZE);
            }

            int key;
            cout << "\nEnter value to search: ";
            cin >> key;

            int result = binarySearch(arr, SIZE, key);

            if(result != -1)
                cout << "Element found at index " << result << " (Position " << result + 1 << ")\n";
            else
                cout << "Element not found in the array.\n";
```

```
            break;
        }

        case 5:
            cout << "\nCurrent Array: ";
            printArray(arr, SIZE);
            break;

        case 6:
            cout << "\nExiting program. \n";
            break;

        default:
            cout << "\nInvalid choice! Please choose between 1 and 6.\n";
    }

} while(choice != 6);

return 0;
}
```

Output:

```
Enter 10 integers for the array:
Element 1: 4
Element 2: 6
Element 3: 8
Element 4: 2
Element 5: 3
Element 6: 1
Element 7: 7
Element 8: 9
Element 9: 7
Element 10: 5

============================
      MENU-DRIVEN PROGRAM
============================
1. Bubble Sort
2. Insertion Sort
3. Selection Sort
4. Binary Search
5. Display Array
6. Exit
Enter your choice (1-6): 2

Array sorted successfully using Insertion Sort!
Sorted array: 1 2 3 4 5 6 7 7 8 9

============================
      MENU-DRIVEN PROGRAM
============================
1. Bubble Sort
2. Insertion Sort
3. Selection Sort
4. Binary Search
5. Display Array
6. Exit
Enter your choice (1-6): 1

Array sorted successfully using Bubble Sort!
Sorted array: 1 2 3 4 5 6 7 7 8 9
```

```
==============================
        MENU-DRIVEN PROGRAM
==============================
1. Bubble Sort
2. Insertion Sort
3. Selection Sort
4. Binary Search
5. Display Array
6. Exit
Enter your choice (1-6): 3

Array sorted successfully using Selection Sort!
Sorted array: 1 2 3 4 5 6 7 7 8 9

==============================
        MENU-DRIVEN PROGRAM
==============================
1. Bubble Sort
2. Insertion Sort
3. Selection Sort
4. Binary Search
5. Display Array
6. Exit
Enter your choice (1-6): 6

Exiting program.

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Process exited after 58.03 seconds with return value 0
Press any key to continue . . .
```